Exponential and Logarithmic Equations

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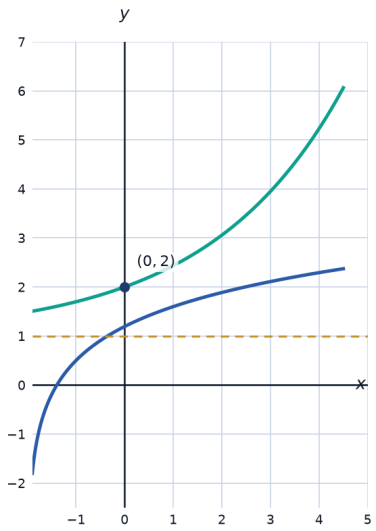
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Exponential equations

Solve $3^{x+1} = 81$ exactly.

2 [Foundational] • Target: Exponential equations

Solve $5^{2x} = 17$. Give an exact answer and a decimal to three places.



3 [Foundational] • Target: Logarithmic equations

Solve $\log_2 x + \log_2 (x - 6) = 4$, checking for extraneous solutions.

4 [Foundational] • Target: Logarithmic equations

The pH of a solution is $\text{pH} = -\log[\text{H}^+]$. Find the hydrogen-ion concentration $[\text{H}^+]$ of a solution with pH 4.6, in scientific notation to two significant figures.

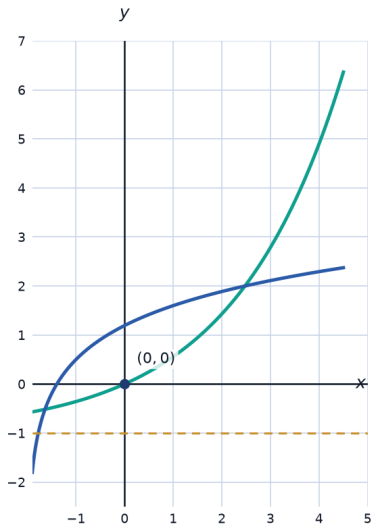
5 [Foundational] • Target: More exponential equations

Solve $2^x = 3^{x-1}$, giving an exact expression and a decimal to three places.

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6 [Foundational] • Target: More exponential equations

Solve $4^x - 3 \cdot 2^x - 4 = 0$ exactly, using the substitution $u = 2^x$.



7 [Foundational] • Target: Systems and applications

A bacteria culture grows according to $N(t) = 500e^{kt}$. If the population reaches 2000 after 6 hours, find k (4 d.p.) and hence the time for the population to reach 10,000, to one decimal place.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Systems and applications

Solve the equation $e^x + e^{-x} = 3$ exactly, using the substitution $u = e^x$ (this expression is related to $2\cosh x$).

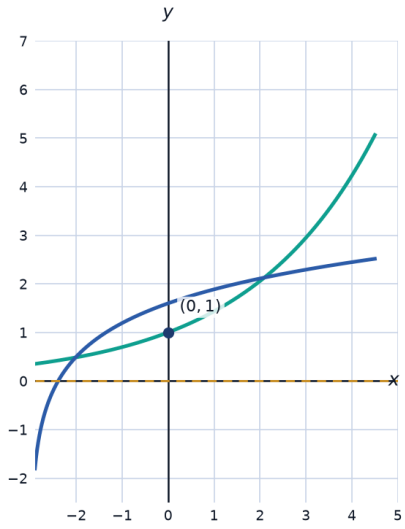
9 [Intermediate] • Target: Combined logarithmic and exponential practice

Solve $9^x = 27^{x-1}$ exactly, writing both sides with base 3.

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10 [Intermediate] • Target: Equations requiring both exponential and logarithmic techniques

Solve $e^{\ln x + \ln 2} = 10$ for x, simplifying the exponent first.



11 [Intermediate] • Target: Verifying solutions numerically

For the equation $5^{2x} = 17$ solved earlier ($x \approx 0.880$), verify the solution by substituting back and computing both sides to 2 decimal places.

12 [Intermediate] • Target: Equations with different bases on each side

Solve $10^{2x-1} = 500$, giving an exact logarithmic expression and a decimal to three places.

13 [Intermediate] • Target: A modeling problem requiring both equation types

A population model is $P(t) = 800e^{0.05t}$. Find the time t when $P(t) = 2000$, to two decimal places, and separately find $P^{-1}(1500)$ (the time when the population reaches 1500), to two decimal places.

14 [Intermediate] • Target: Checking solutions by substitution

Verify the solution $x = 8$ found earlier for $\log_2 x + \log_2 (x - 6) = 4$ by substituting back into the original equation.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

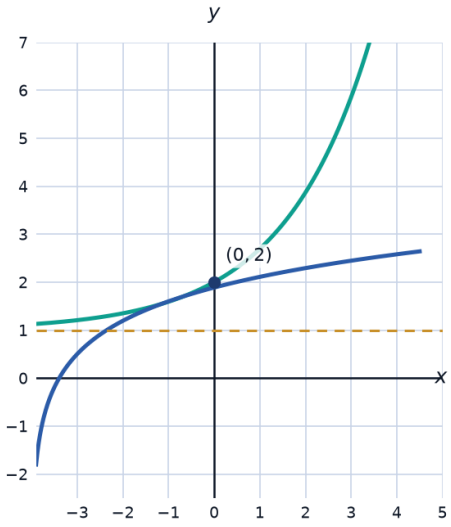
15 [Challenging] • Target: Checking solutions by substitution • Cross-Lesson: 3.01 The Unit Circle and Exact Values

A radioactive sample decays according to $A(t) = A_0 e^{-0.08t}$. If the initial amount is 500 g, find the time t (to two decimal places) at which only 50 g remains, and separately find the time at which 90% of the original sample has decayed away.

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16 [Challenging] • Target: Equations requiring both exponential and logarithmic techniques • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Level 2 · MediumSolve the logarithmic equation $\log_2(x+3) + \log_2(x-1) = 5$ analytically for all real solutions. Show the verification step that eliminates extraneous roots.



17 [Challenging] • Target: Verifying solutions numerically • Cross-Lesson: 3.03 Sum and Difference Formulas

Find all real solutions to the rational equation $\frac{x}{x-2} + \frac{1}{x+3} = \frac{5}{x^2+x-6}$.

18 [Challenging] • Target: Equations with different bases on each side • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Solve the exponential equation $4^{x+1} - 5(2^x) + 1 = 0$ for all real values of x.

19 [Challenging] • Target: A modeling problem requiring both equation types • Cross-Lesson: 4.01 Circles and Parabolas

Solve the logarithmic equation $\log_4(x+6) + \log_4(x-6) = 3$, checking for extraneous roots.

20 [Challenging] • Target: Checking solutions by substitution • Cross-Lesson: 4.02 Ellipses

[Exponential & Logarithmic Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Exponential equations

What is the sum of all real solutions to the equation $2^{\log_5(x^2)} - 6 \cdot x^{\log_5(2)} + 8 = 0$?

a. 30

b. 26

c. 6

d. 3

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22 [Foundational] • Target: Logarithmic equations

The rate of change of a quantity N is proportional to its current value, modeled by the differential equation $\frac{dN}{dt} = kN$, where k is a constant. If the initial population at time $t = 0$ is N_0, and at time $t = 5$ years, the population has grown to $2N_0$, what is the population at time $t = 15$ years, expressed in terms of N_0?

a. $6N_0$

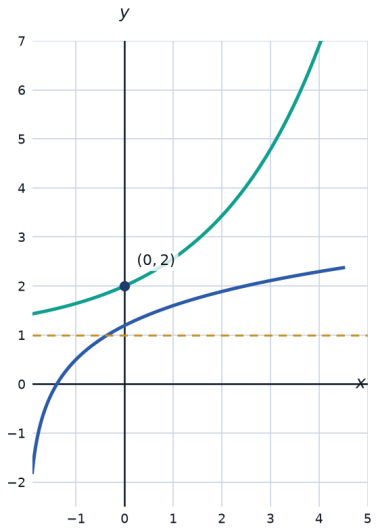
b. $8N_0$

c. $7N_0$

d. $10N_0$

23 [Foundational] • Target: More exponential equations

The equation $\log_a x = b$ and $\log_b x = a$ simultaneously implies:



a. No simultaneous solution

b. $a = b$

c. $x = 0$

d. $x = a^b = b^a$

24 [Foundational] • Target: Systems and applications

Find the number of solutions to $e^x = x + 2$.

a. 1

b. 2

c. Infinitely many

d. 0

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25 [Foundational] • Target: Combined logarithmic and exponential practice

Solve $e^x = x + 2$ (number of solutions).

a. 0 solutions

b. 2 solutions (one for $x > 0$, one for $x < 0$)

c. 1 solution

d. Infinitely many

26 [Foundational] • Target: Equations requiring both exponential and logarithmic techniques

Find all solutions to $4^x - 6 \cdot 2^x + 8 = 0$.

a. $x = 0, 1$

b. $x = 2, 4$

c. $x = 3, 4$

d. $x = 1, 2$

27 [Foundational] • Target: Verifying solutions numerically

Find the number of real solutions to $\ln x = x^2 - 3$.

a. 3

b. 1

c. 0

d. 2

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28 [Foundational] • Target: Equations with different bases on each side

The graph of the function $f(x) = a \cdot 2^{x-h} + k$ is shown below. The graph passes through the points $(2, 9)$ and $(4, 33)$. The horizontal asymptote is $y = 1$. Find the value of a.



a. 4

b. 2

c. 1/2

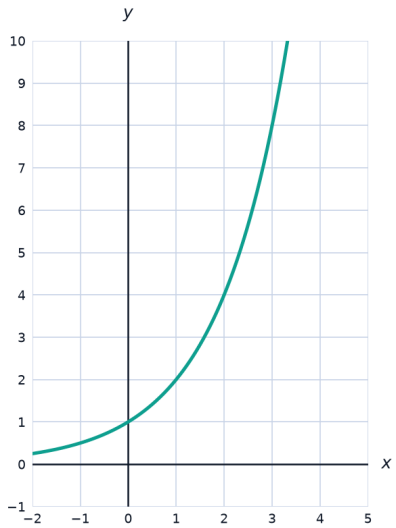
d. 8

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: A modeling problem requiring both equation types**

Consider the function $f(x) = 2^x$. Which of the following points lies on the graph of $f(x)$?



a. $(1, 1)$

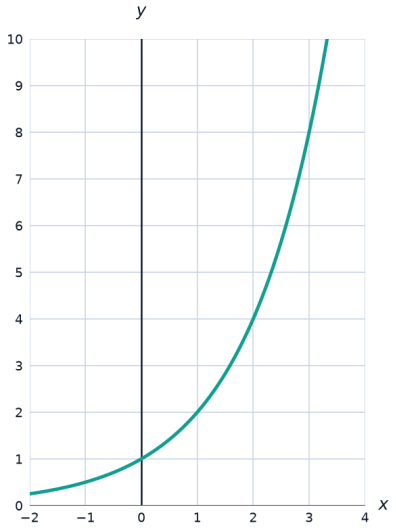
b. $(2, 4)$

c. $(3, 6)$

d. $(0, 2)$

30 **[Intermediate]** • **Target: Checking solutions by substitution**

The graph of the exponential function $f(x) = 2^x$ is given. A point P on the graph has a y-coordinate of 8. What is the x-coordinate of point P?



a. 1

b. 2

c. 3

d. 4

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31 **[Intermediate]** • **Target: Exponential equations**

The function f is defined for $x \in \mathbb{R}$ by $f(x) = 3x + k,$ where k is a constant. The function g is defined for $x \in \mathbb{R}$ by $g(x) = |x - 4|.$ Given that $f^{-1}(4) = 2$, find the value of k and all real solutions to the equation $f^{-1}(x) = g(x)$.

a. $k = -2;\ x = \frac{5}{2},\ x = 7$

b. $k = -2;$ \; x = 7\)

c. $k = 2;\ x = \frac{7}{2},\ x = 5$

d. $k = -2;\ x = -\frac{5}{2},\ x = 7$

32 **[Intermediate]** • **Target: Logarithmic equations**

The bar graph shows the distribution of the number of goals scored per game by a soccer team across all 40 games of its regular season. If the team plays k post-season games and scores exactly 3 goals in each of those games, the mean number of goals scored per game across all regular-season and post-season games combined will be exactly 2. What is the value of k?

a. 8

b. 12

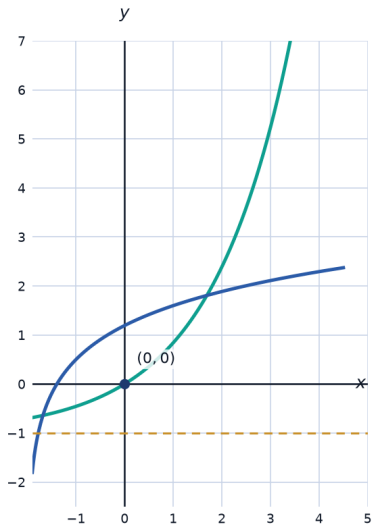
c. 20

d. 24

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33 [Intermediate] • Target: More exponential equations

Consider the equation $\sqrt[3]{25 - x^2} + \sqrt[3]{x^2 + 3} = 4$ What is the product of all real solutions to this equation?



a. -24

b. 24

c. -48

d. 0

34 [Intermediate] • Target: Systems and applications

Consider the function $f(x) = \log_3(ax + b)$, where a and b are positive constants. If the graph of $y = f(x)$ passes through the points $(1, 2)$ and $(10, 3)$, find the value of a.

a. 3

b. 1

c. 1/3

d. 9

35 [Intermediate] • Target: Combined logarithmic and exponential practice

What is the horizontal asymptote of the graph of the function $f(x) = 3 \cdot 2^{x+1} - 5$?

a. $y = -5$

b. $y = 0$

c. $y = 3$

d. $x = -1$

36 [Intermediate] • Target: Equations requiring both exponential and logarithmic techniques

The graph of the function $g(x) = a \cdot b^x$ passes through the points $(1, 6)$ and $(3, 24)$. If a and b are positive constants, what is the value of a?

a. 1.5

b. 2

c. 3

d. 6

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37 [Intermediate] • Target: Verifying solutions numerically

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a and the value of b.

a. a = 4, b = 3

b. a = 3, b = 4

c. a = 6, b = 2

d. a = 2, b = 6

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Equations with different bases on each side • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a.

a. 2

b. 3

c. 4

d. 6

39 [Challenging] • Target: A modeling problem requiring both equation types • Cross-Lesson: 4.01 Circles and Parabolas

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(2, 36)$. If $b > 0$ and $beq1$, find the value of a.

a. 4

b. 3

c. 12

d. 1

40 [Challenging] • Target: Checking solutions by substitution • Cross-Lesson: 4.02 Ellipses

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 6)$ and $(2, 18)$. What is the value of b?

a. 2

b. 3

c. 4

d. 6

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41 [Challenging] • Target: Exponential equations • Cross-Lesson: 4.03 Hyperbolas

The graph of the exponential function $f(x) = a \cdot 2^x$ passes through the point $(3, 40)$. What is the value of a?

a. $\frac{5}{2}$

b. 5

c. 10

d. 20

42 [Challenging] • Target: Logarithmic equations • Cross-Lesson: 5.01 Vectors: Components and Operations

Consider the matrix equation: $(3, -1; 2, k)(x; y) = (5; 7)$. If the matrix $(3, -1; 2, k)$ is singular, what is the value of k?

a. $-3/2$

b. $3/2$

c. $-2/3$

d. $2/3$

43 [Challenging] • Target: More exponential equations • Cross-Lesson: 5.02 The Dot Product

What is the value of the exponential function $f(x) = 2^x$ when $x = 3$?

a. 6

b. 8

c. 9

d. 5

44 [Challenging] • Target: Systems and applications • Cross-Lesson: 5.03 Parametric Equations

What is the value of y when $x = 3$ for the function $f(x) = 2^{x+1}$?

a. 4

b. 8

c. 6

d. 16

45 [Challenging] • Target: Combined logarithmic and exponential practice • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Solve: $e^{2x} = 5$. Express in terms of ln.

a. $x = \ln k**2 + 1 - k$

b. $x = \ln \frac{k**2 + 1}{k}$

c. $x = \frac{\ln k**2 + 1}{k}$

d. $x = \frac{k}{\ln k**2 + 1}$

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